Syed Sulaiman

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Objective

Embedded Circuit Design Engineer with hands-on Experience in PCB design, embedded systems and vehicle electronics across Formula Student and robotics applications. Interested in developing reliable, high-performance electronic systems through strong fundamentals in system architecture and validation

Education

- **PES University** *2021 - 2025*
  B.Tech in Electronics and Communications, ***GPA: 8.77/10.00*** Bangalore, India
  - Specialization in Signal Processing and Systems Engineering
  - Minors in Computer Science Engineering
- **Christ Junior College** *2021*
  Pre-University Education, ***Grade: 95.6%*** Bangalore, India
- **St. Paul's English School** *2019*
  Secondary Education, ***Grade: 94.83%*** Bangalore, India

Experience

- **Systemantics India Pvt Ltd []** *January 2025 - Present*
  Associate Design Engineer - Robotics Bangalore, India
  - Designed and developed embedded electronics subsystems for a collaborative robot, including Raspberry Pi(CM5) based system control, stepper motor controllers, force-torque sensing modules, and LVDT controllers, with end-to-end schematic and PCB design following DFM/DFA principles.
  - Performed circuit design, power budgeting, component selection and tolerance analysis, including buck/boost power supply design with detailed calculations and worst-case analysis; validated through testing and debugging.
  - Designed STM32-based embedded hardware featuring CAN, SPI, I²C, UART, RS485, ADC/DAC interfaces.
  - Worked on high-speed interfaces including Gigabit Ethernet, USB 3.0, HDMI, and PCIe, considering signal integrity, routing constraints, and PCB stack-up design.
  - Conducted hardware validation and debugging using oscilloscopes and logic analyzers, performing root cause analysis(RCA) to ensure robust performance, reliability, and system integration.
  - Led the development of a jet engine turbine blade metrology and inspection station, integrating LVDT-based dimensional measurement with a machine vision diffuse dome illumination system for automated defect detection.
  - Developed STM32-based embedded firmware for real-time control, communication, and hardware interfacing across multiple subsystems, including closed-loop stepper motor servo control with encoder feedback
- **Artificial Intelligence and Robotics Lab (AIRL), Indian Institute of Science** *June 2024 - August 2024*
  Robotics Intern Bangalore, India
  - Set up and configured a Unmanned Underwater Vehicles (UUVs) ROS–Gazebo simulation environment; conducted a literature review on swarm coordination and developed a multi-agent distributed area-search simulation.
- **Systemantics India Pvt Ltd []** *June 2023 - July 2023*
  Design Intern Bangalore, India
  - Conducted a market analysis and technical evaluation of 10+ depth camera systems to identify a suitable wrist-mounted vision system for a collaborative robot arm
  - Designed custom electronics and PCBs for the vision system, including a Raspberry Pi Camera HAT with a ToF distance and LED flash support board
  - Developed STM32 firmware for a stepper motor driver, implementing low-level motor control routines for integration with the robot's gripper mechanism.

Leadership Experience

- **Electrical Lead, Formula Student team** *2022 - 2025*
  Vega Racing Electric, PES University []
  - Played a key role in rebuilding the team and developing a fully functional race car from scratch; Leading end-to-end development of the vehicle's electrical architecture (high-voltage and low-voltage), including energy storage, safety systems, and embedded subsystems for an electric race car.
  - Designed the traction system energy storage from scratch, ensuring compliance with strict Formula Student safety regulations related to accumulator structural integrity, cell monitoring, isolation and high-voltage protection.
  - Performed system-level DFMEA systematically analysing failure modes across HV/LV systems, safety circuits (shutdown logic) and system-critical signals to ensure rule compliance and safe operation.
  - Led system-level validation and debugging using RCA across battery management, motor control and VCU.

- Architected and implemented an ESP32-based vehicle data acquisition and telemetry system integrating 10+ sensors and CAN communication with BMS and motor controller, enabling real-time monitoring, onboard logging, and driver visualization via GUI and GPS-enabled ground station.
- Established structured design reviews and cross-functional integration workflows with mechanical, firmware, and systems teams to ensure robust and manufacturable designs.
- Delivered a fully compliant and validated electrical system, successfully passing strict Formula Student technical inspections on first attempt.

PROJECTS

- **Formula Student Electric Vehicle – Low Voltage Architecture & Safety-Critical Circuits**
  Tools: KiCAD, EasyEDA, LTSpice, MultiSim, Oscilloscopes & Logic Analyzers
  - Architected and implemented the complete low-voltage electrical system with load analysis and worst-case design calculations with strong focus on failure mode identification and mitigation to ensure safe, fault-tolerant operation.
  - Designed and validated standalone analog safety-critical circuits (no MCU dependency) including:
    - Shutdown Circuit (SDC) for immediate HV system deactivation under fault conditions
    - Brake System Plausibility Device (BSPD) to detect unintended motor torque during braking
    - Tractive System Active Light (TSAL) for high-voltage state indication
    - Precharge and interlock circuits for controlled and safe high-voltage system activation
    - Insulation monitoring interface to ensure safe isolation between High and Low-voltage domains
  - Developed multiple PCBs using discrete analog design (comparators, op-amps, MOSFETs, relays), incorporating tolerance analysis, component derating, grounding strategy, and EMI/noise mitigation.
  - Designed HV–LV interfacing and ensured strict compliance with Formula Student rules, including isolation requirements, System Critical Signal handling, and use of safety-rated components (e.g., UL94-V0 for HV paths).
  - Built and validated complete electrical integration including wiring harness design, connector selection, and crimping standards, ensuring robustness in high-vibration environments.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[J.1] **Sulaiman, S**. et al. **WareOdo: Leveraging warehouse landmarks for correcting odometry in indoor mobile robots**. *Intel Serv Robotics*, Vol. 18, pp. 687–708 (2025). DOI: https://doi.org/10.1007/s11370-025-00610-4

- Collaborated with Accio Robotics to develop and validate a landmark-assisted odometry correction framework using EKF-based sensor fusion and AprilTag based global pose correction, achieving 60% drift reduction and state-of-the-art localization accuracy across simulated and real world warehouse environments.

SKILLS

- **Electronics Software Tools:** KiCad, Autodesk Eagle, EasyEDA, Altium Designer, LTspice
- **Embedded programming Tools:** STM32CubeIDE, Arduino IDE, FreeRTOS
- **Communication Protocols:** CAN, I2C, SPI, USART, USB, Ethernet, Modbus
- **Modeling & Simulation Tools:** MATLAB, Simulink
- **Programming Languages:** C, C++ , Embedded C, Python, Shell scripting, Java
- **Computer Vision:** OpenCV, PyTorch, TensorFlow, Google Cloud Vision, YOLO
- **Robotics Tools:** ROS, RViz, Gazebo, MoveIt
- **CAD:** Fusion 360

HONORS AND AWARDS

- **2nd Prize, HackeZee** — *October 2022*
  Hardware Design Contest, Dept. of ECE [🌐]
  - Secured Second Prize among 78 teams; awarded a cash prize for the project.
  - Designed and built an Omnidirectional Robot Platform with 3D-printed Mecanum wheels, enabling full 360° motion; implemented multi-mode control (Bluetooth, web interface, RF) and programmed onboard logic for recording and replaying motion sequences, demonstrating robust embedded control automation capabilities.
  - Demonstrated skills in circuit and PCB design, embedded programming, motor control, wireless communication and rapid hardware prototyping.
- **2nd Place, Robofest 2.0** — *November 2022*
  24hr hackathon, IEEE RAS,PES University [🌐]
  - Developed and deployed a ROS-based perception and manipulation pipeline for a 6-DOF serial manipulator, integrating OpenCV-based object detection to sort objects by shape and color; successfully demonstrated in both RViz/Gazebo simulation and on a physical robotic platform.
  - Demonstrated skills in Robot Simulation, Computer Vision, inverse kinematics and servo control